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ACS Appl. Mater. Interfaces, **Just Accepted Manuscript** • DOI: 10.1021/acsami.5b07909 • Publication Date (Web): 26 Oct 2015Downloaded from <http://pubs.acs.org> on October 31, 2015**Just Accepted**

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Engineering the Electrochemical Capacitive Properties of Micro-Supercapacitors Based on Graphene Quantum Dots/MnO₂ using Ionic Liquid Gel Electrolytes

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ABSTRACT

All-solid-state micro-supercapacitors (MSCs) have been receiving intense interest due to their potential as micro/nano-scale energy storage devices, but their low energy density has limited practical applications. It has been reported that gel electrolytes based on ionic liquids (ionogels) with large potential windows can be used as solid electrolytes to enhance the energy density of MSCs, but a systematic study on how to select and evaluate such ionogels for MSCs is rare. In this study, we construct a series of all-solid-state asymmetric MSCs on the interdigital finger electrodes, using graphene quantum dots (GQDs) as the negative electrode, MnO₂ nanosheets as the positive electrode and different ionogels as the solid electrolytes. Among them, the MSC using 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide ([EMIM][NTF₂]) with 4 wt% fumed SiO₂ ionogel exhibited the best electrochemical performance, having excellent rate capability with the scan rate up to 2000 V s⁻¹, ultrafast

frequency response ($\tau_0=206.9 \mu\text{s}$) and high energy density. The outstanding performance of this device mainly results from fast ion diffusion, high ion conductivity of the ionogel, and ionic liquid–matrix interactions. The results presented here provide guidance for picking out appropriate ionogels for use in high-performance all-solid-state MSCs to meet the growing requirement of micro-nanoscale energy storage devices. Additionally, the ultrafast frequency response of our MSCs suggests potential applications in ac line-filters.

KEYWORDS: Micro-supercapacitors, graphene quantum dots, ionic liquid gel, rate capability, AC line-filters, mechanism

1. INTRODUCTION

The recent technological trend towards miniaturized electronic devices, such as nanorobotics, micro-electromechanical systems (MEMS), and epidermal and implantable medical devices, has greatly stimulated the need for small-scale energy storage systems.¹⁻² Micro/nanoscale energy storage units are ideal for these types of applications and can be made small enough to be directly integrated into electronic circuits.³ Alternatively, micro-supercapacitors (MSCs) can serve as new miniaturized energy-storage devices on the basis of their advantages (high power density, fast rate capability, long cycling lifetime and high-frequency response),⁴⁻⁵ and as such are crucial for special applications, such as for ac line-filtering.⁶⁻⁸

Recently, considerable efforts have been made towards enhancing the performance of MSCs, mainly by optimizing device architecture and improving electrode materials. With respect to device architecture, interdigitated planar MSCs have been widely investigated due to their short ion-path and improved kinetic performance, and more importantly because such on-chip

MSCs are more easily integrated into electronic circuits.⁹ Research on electrode materials has focused on the development of nanostructured materials. Among these, graphene based materials, such as multilayer graphene,¹⁰⁻¹⁴ doped graphene,¹⁵ graphene/CNTs,¹⁶⁻¹⁸ graphene quantum dots,^{5,19} graphene/polyaniline,^{3,20} graphene/MnO₂²¹ and graphene/MnO₂/silver nanowire,²² have been explored most extensively as active materials for planar MSCs. Further investigations have been made on asymmetric MSCs which combine a capacitor-type electrode and a battery-type Faradaic electrode to enhance capacitance and energy density.²³⁻²⁶ Despite these efforts, the energy densities of most MSCs are still insufficient to meet the requirements of micro-devices.

According to the equation $E = \frac{1}{2}CV^2$,²⁷ the energy density of supercapacitors (SCs) is proportional to its capacitance and to the square of its operating potential window (OPW). Accordingly, increasing the OPW of SCs is a more efficient method to increase energy density than increasing its capacitance.²⁸ Ionic liquids (ILs) have been recently recognized as promising candidates to replace conventional electrolytes in electric double-layer capacitors (EDLCs) owing to their wide OPW, high ionic conductivity, good thermal stability, and non-volatility.²⁹ Compared with liquid electrolytes, all-solid-state SCs have many advantages, such as safety (without leakage of liquid electrolytes), environmental friendliness, stability, and portability.³⁰ Aqueous gel electrolytes have been extensively applied in the MSCs,^{3,12-21,23,25-27,31} and while there has been relatively little research on using ionogels in MSCs, including 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide ([BMIM][NTF₂]) fumed silica (FS) nanopowder,^{11,22} 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide ([EMIM][NTF₂]) FS nanopowder,³² 1-ethyl-3-methylimidazolium tetrafluoroborate ([EMIM][BF₄]) FS nanopowder,³³ and [EMIM][NTF₂]-poly (ethylene glycol) diacrylate.³⁴

However, these works mainly focus on device architecture and the preparation of electrode materials, and studies on how to select and evaluate ionogels are very limited for MSC systems. Therefore, a systematic study of ionogel electrolytes is necessary, including the matching between electrode materials and ionogels, the interaction mechanism between the solid matrix (e.g. polymer, nanoparticles) and the ILs, and finally, the influence on electrochemical characteristics. To the best of our knowledge, research in this area is rare.

We have previously reported MSCs based on graphene quantum dots (GQDs) which exhibit excellent capacitive performance in aqueous electrolytes.^{5,35} However, the areal capacity and energy density of these MSCs need to be further enhanced. In this work, all-solid-state asymmetric MSCs based on GQDs//MnO₂ were firstly built via controllable electrodeposition, and the effects of different ionogels on the electrochemical behavior of GQDs//MnO₂ MSCs were examined. The electrochemical performance of the MSCs based on EMIM-NTF₂-4 wt% FS is the best, presenting the highest energy density and fastest frequency response than that in other ionogels. This result can be explained by ion diffusion, ion conductivity, and IL-matrix interactions. The fundamental understanding will provide guidance for selecting appropriate ionogels and designing high-performance ionogel-MSCs. These MSCs in particular have potential applications in ac line-filters.

2. EXPERIMENTAL

2.1 Materials and chemicals

Two kinds of ILs, [EMIM][NTF₂] and [EMIM][BF₄], were obtained from the Center for Green Chemistry and Catalysis, Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences. All ILs were vacuum-dried at 110 °C overnight before use. Poly (vinylidene fluoride-co-hexafluoropropylene) (PVDF-HFP, average Mw ~400,000 and average Mn ~130,000) and

fumed silica nanopowder (FS, average diameter ~ 7 nm) was purchased from Sigma-Aldrich. High-purity graphite powder (99.9%, 325 mesh) was purchased from Qingdao Huatai Tech. Co., Ltd. N, N-dimethylformamide (DMF) was purchased from Tianjin Reagent Company. All conventional chemicals were of analytical grade and were used without further purification. Ultrapure water ($18 \text{ M}\Omega \text{ cm}^{-1}$) was used throughout the experiments.

2.2 Fabrication of GQDs

Graphene oxide (GO) sheets were synthesized from natural graphite using a modified Hummers' method.³⁶ GQDs were generated from the GO powder by a facile one-step solvothermal method according to our previous work.^{5,35} In a typical synthesis, as-made GO powder (540 mg) is suspended in DMF (40 mL) and sonicated for 30 min. The suspension is then transferred to a poly (tetrafluoroethylene) (Teflon)-lined autoclave (50 mL) and heated at 200°C for 8 h. After that, the mixture is filtered using a $0.22 \mu\text{m}$ microporous membrane and the brown filter solution is collected, that is the GQDs dispersion in DMF. Dry GQDs powder is obtained from the GQDs suspension by removing DMF using a rotary evaporator.

2.3 Preparation of GQDs// MnO_2 asymmetric MSCs

Before the electrochemical deposition, the interdigitated Au electrode are immersed in acetone, ethanol and isopropanol for more than 20 min, successively. First, electro-deposition of the MnO_2 anode on one side of the interdigitated Au electrode was performed at a constant current of 2 mA cm^{-2} with the potential window of -1.2 V to 1.2 V for 5 min in a solution of 0.01 M manganese acetate (MnAc_2) and 0.02 M ammonium acetate (NH_4Ac) containing the volume fraction of 10% dimethyl sulfoxide (DMSO) and 90% ultrapure water under ambient temperature. A platinum sheet and an Ag/AgCl electrode were used as the counter and the reference electrodes, respectively. Subsequently the electrodes were washed with ultrapure water

and dried in air. GQDs were then electrodeposited on another side of the interdigitated Au electrode in DMF solution (50 mL) containing GQDs (3.0 mg) and $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (10 mg) at a constant voltage of 80 V for a duration of 30 min using electrophoretic deposition (EPD) technology (Keithley 2400 Sourcemeter). Finally, the electrodes were again washed with ultrapure water and then were placed in drying oven for more than 6 h at 60 °C. The schematic illustration of two-step electro-deposition on an interdigital finger electrode is shown in Fig. S1e.

2.4 Preparation of ionogels

For the preparation of ionogels based on PVDF-HFP, 0.6 g of PVDF-HFP was added to 6 mL of acetone and then stirred magnetically for more than 4 hours until the solution became transparent, subsequently 2.4 g of $[\text{EMIM}][\text{BF}_4]$ and $[\text{EMIM}][\text{NTF}_2]$ were added into the above solution, respectively and stirred in a glove box under an Ar atmosphere for more than 2 hours, resulting in EMIM- BF_4 -20 wt% PVDF-HFP and EMIM- NTF_2 -20 wt% PVDF-HFP ionogels. For the preparation of ionogels based on FS nanopowder, 2 g of $[\text{EMIM}][\text{BF}_4]$ and 0.5 g of FS nanopowder were stirred in a glove box under an Ar atmosphere for more than 6 hours to get the clear viscous ionogel, EMIM- BF_4 -20 wt% FS. The fourth type of ionogel was prepared by mixing together $[\text{EMIM}][\text{NTF}_2]$ with different contents of FS nanopowder and stirring in a glove box for more than 12 hours to obtain the viscous ionogels of EMIM- NTF_2 -3 wt%, 4 wt%, 5 wt%, 6 wt% FS.

2.5 Characterization and Electrochemical measurements

FESEM (JSM-6701F) and TEM (JEOL 2100 FEG) were performed to characterize the morphologies of the as-prepared samples. AFM was conducted on a Digital Instrument Nanoscope IIIa AFM (Veeco) in tapping mode. XRD patterns were measured on a powder X-ray diffraction system (XRD, TTR-III) equipped with Cu K α radiation ($\lambda=0.15406$ nm). Raman

spectra were recorded on a JY-HR800 Raman spectrometer using a laser excitation wavelength of 532 nm. The thermal stability of the ionogels was characterized by thermogravimetric analysis (STA449C) at a rate of 10 °C min⁻¹ under N₂ atmosphere. Differential scanning calorimetry (200F3) was carried out at a heating rate (10 °C min⁻¹) under N₂ from -100 to 20 °C. Infrared spectra were recorded on an IFS120HR FTIR spectrometer. PL emission spectra were obtained using an F-4500 fluorescence spectrophotometer. The electrical conductivity of the ILs was measured with a conductivity meter (Mettler, Toledo FE30 FiveEasy). X-ray photoelectron spectroscopy (XPS) analyses were carried out on an X-ray photoelectron spectrometer (ESCALAB250xi) with an Al K Alpha as the X-ray source. All electrochemical tests, including cyclic voltammetry (CV), galvanostatic charge/discharge (GCD), and electrochemical impedance spectroscopy (EIS) were carried out in a two-electrode system using an electrochemical workstation (CHI660D, Shanghai, China) in the glove box. Ionic conductivity was measured in the thermostated device (KW-TH-150X) made by Dongguan KOWZN testing equipment Co. Ltd.

The specific capacitance was calculated from the GCD curves based on Eq. (1):

$$C = \frac{I\Delta t}{A\Delta V} \text{ (}\mu\text{F cm}^{-2}\text{)} \quad (1)$$

The specific energy density and power density are defined according to Eq. (2) and (3), respectively:

$$E = \frac{C\Delta V}{7200} \text{ (}\mu\text{Wh cm}^{-2}\text{)} \quad (2)$$

$$P = \frac{E \times 3600}{\Delta t} \text{ (}\mu\text{W cm}^{-2}\text{)} \quad (3)$$

Where, I is the current density (μA), Δt is the discharge time (s), ΔV is the potential window of the discharging (V), A is the geometric surface of substrate, E is the energy density ($\mu\text{Wh cm}^{-2}$),

and P is the energy density ($\mu\text{W cm}^{-2}$).

2.6 Ionic Conductivity Measurements

The ionic conductivity was determined using a homemade cell by means of impedance using CHI660D at an AC amplitude of 10 mV. The frequency range employed for the measurements varied from 100 kHz to 0.05 Hz. The cell was made of two 1 cm stainless steel quadrate plates (electrodes) fixed to a PTFE sheet with conductive lug, then placed into conformal PTFE container with a groove and sealed. For the polymer electrolytes, a film was sandwiched between mirror-finished stainless steel electrodes, sealed in a Teflon container, and subjected to the impedance measurements. For the colloidal ionogels, EMIM-NTF₂-FS was filled in the cell and the thickness was obtained by measuring the thickness of the entire cell before and after the formation of the ionogel, the cell was then placed in a thermostated device. The temperature was controlled to within ± 0.1 °C of the set point. All ionogels were thermally equilibrated at each temperature for 10 min prior to the measurements. Each measurement was conducted three times and the mean value was calculated and reported.

The conductivity values (σ) are calculated from Eq. (4):³⁷

$$\sigma = \frac{1}{R_s} \frac{D}{A} \quad (\mu\text{F cm}^{-2}) \quad (4)$$

Where, R_s is the equivalent series resistance, D is the thickness of the ionogel, and A is the contact area between the stainless steel plate and the ionogel.

3. RESULTS AND DISCUSSION

3.1 Structure characterizations and construction of GQDs//MnO₂ asymmetric MSCs

The TEM image of the as-prepared GQDs (Fig. 1a) shows that GQDs are monodisperse, having a relatively narrow lateral size distribution ranging from 1 nm to 6 nm with an average

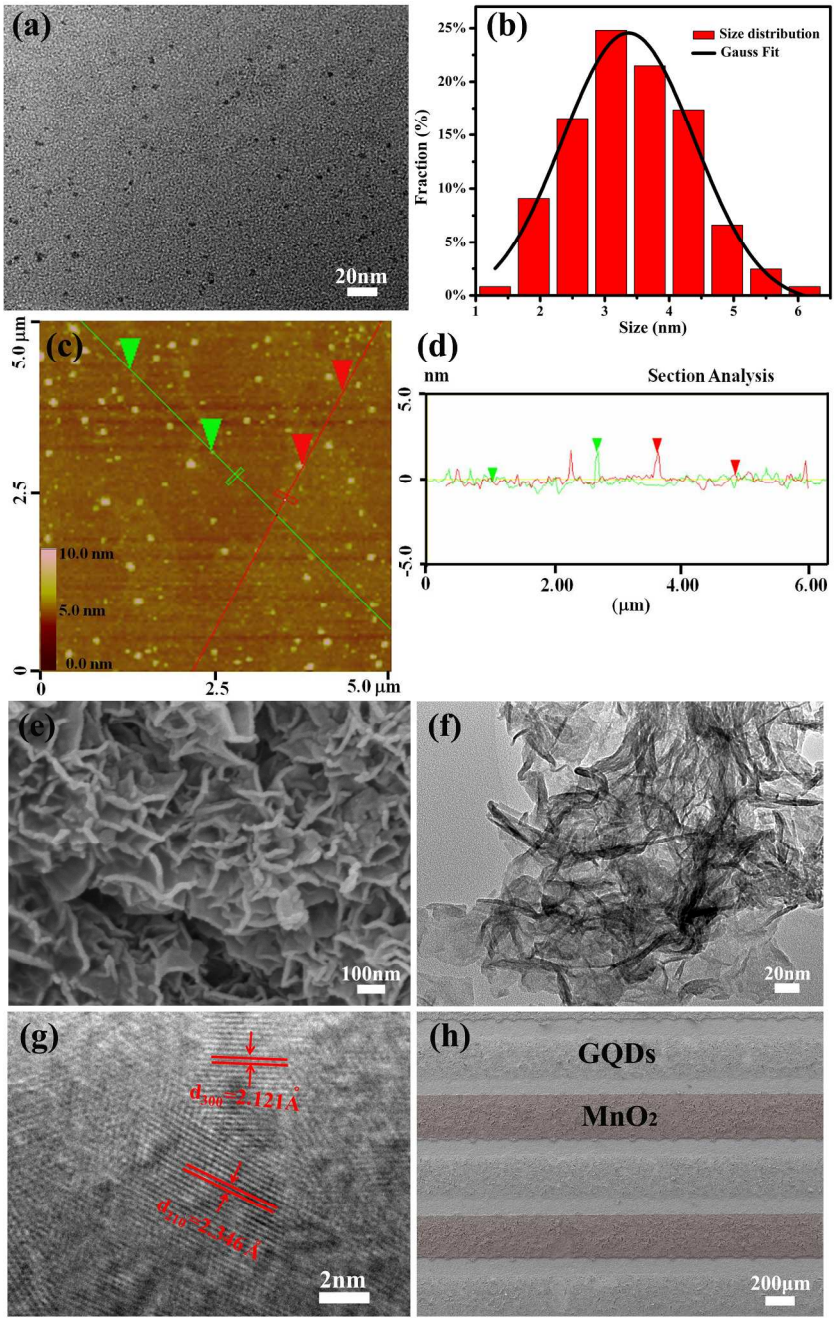


Figure 1. a) TEM image of as-prepared GQDs, b) Size distribution of GQDs, c) AFM image of GQDs on a Si substrate, d) Height profiles along the green and red lines in (c); e, f) SEM and TEM images of electrodeposited MnO₂ respectively, g) high-resolution TEM image of as-prepared MnO₂, h) low-magnification SEM dyeing image of the interdigital finger electrodes after deposition of GQDs and MnO₂.

diameter of 3.38 nm (Fig. 1b). High-resolution TEM (Fig. S1a) demonstrates the 0.243 nm interlayer spacing of the GQDs, representing the (1120) lattice fringe of graphene.³⁵ The AFM image of the GQDs shown in Fig. 1c demonstrates the topographic morphology of the GQDs, which have heights between 0.5 nm and 2.8 nm (Fig. 1d). The formation of GQDs is also confirmed by photoluminescence (PL) spectra (Fig. S1b) which shows a strongest peak at 423 nm when excited at 340 nm, presenting an excitation-dependent feature. A solution of GQDs in DMF excited by a 365 nm lamp (6 W) exhibits an intense blue-green luminescence (Fig. S1c), and the GQDs solution remains homogeneous even after one year at room temperature without any perceptible change. Fig. 1e and f show the as-deposited MnO₂ electrode consists of ultrathin nanosheets with a lateral dimension of 100-300 nm. Moreover, a large number of nanopores are present among these nanosheets, which are beneficial to electron transport and reduce the diffusion path of ions. The XRD pattern (Fig. S2d) of the MnO₂ electrode indicates that the obtained MnO₂ is γ -MnO₂, and the high-resolution TEM (HR-TEM) further confirms this. As shown in Fig. 1h, the lightly coloured region in the SEM image represent the GQD cathode, while the darker region indicate the MnO₂ anode.

The fabrication of the GQDs//MnO₂ asymmetric MSCs can be carried out via a controllable two-step electro-deposition on an interdigital finger electrode composed of 32 in-plane interdigitated Au microelectrodes (16 positive and 16 negative microelectrodes). Each microelectrode is 230 μ m wide and 10 mm in length, and the distance between adjacent microelectrodes is 200 μ m. As shown in Fig. S1e, electro-deposition of MnO₂ using an electrochemical work station is done to fabricate the cathode, and the loading mass of MnO₂ positive electrode materials is about 0.08 mg. The electro-deposition of the cathode material is done at a constant current of 2 mA cm⁻² with the potential window of -1.2 V to 1.2 V (Fig. S2b).

The deposition of GQDs on another side of the Au electrode is performed using EPD technology from a DMF suspension of GQDs containing $\text{Mg}(\text{NO}_3)_2$. Fig. S2a shows the constant-voltage EPD process which is run at the voltage of 80V for 1800 s, over which the deposition current sharply decreases, especially during the first 300 s as seen in Fig. S2a, and the loading mass of GQDs negative electrode materials is about 0.2 mg.

Electrochemical properties of the GQDs// MnO_2 asymmetric MSCs could be determined by both of cathode and anode on the interdigitated Au microelectrodes. Firstly, we study the electrochemical properties of each anode and cathode. As shown in Fig. S3a, b, the CV curves of GQDs single electrode exhibit a regular rectangle in sharp and a very rapid current response even at a high scan rate of 100 V s^{-1} , indicating fast charge transfer and diffusion. The specific capacitance of the GQDs single electrode is calculated to be 0.86 mF cm^{-2} according to GCD curves at the current density of $136 \mu\text{A cm}^{-2}$. The EIS plot is nearly perpendicular to the real axis in the low-frequency region (Fig. S3d), revealing an ideal capacitive behavior. Correspondingly, Fig. S4 shows the electrochemical properties of MnO_2 single electrode. The CV curves of the MnO_2 single electrode maintain a nearly rectangular CV shape even at a high scan rate of 10 V s^{-1} . It can be observed that all the charge-discharge curves are linear and symmetrical at different current densities shown in Fig. S4c, suggesting a high coulombic efficiency. The semicircle at high frequency is almost invisible according to the EIS plots of the MnO_2 single electrode, presenting very small charge-transfer resistance (R_{ct}) and small contact resistance resulting from strong adhesion between MnO_2 and Au substrate.

As regards the electrochemical window, it is determined by not only the electrochemical stability window of the electrolyte, but also the potential window of the anode and cathode materials. The potential window of GQDs electrode ranges from -0.5 V to 1.0 V, and MnO_2

electrode ranges from 0 V to 1.8 V. So the electrochemical window of GQDs//MnO₂ asymmetric MSCs is no higher than 2.5 V. We also tested different potential windows (from 2.2 V to 3 V) of the asymmetric MSCs in EMIM-NTF₂-Fumed SiO₂ as shown in Fig. S5b. The results shows that the electrochemical window of asymmetric MSC can reach up to 2.5 V, and the polarization phenomenon had been obvious gradually when the electrochemical window of the asymmetric MSCs is more than 2.5 V. Although the ionogels preserve the main properties of ILs, the ionic conductivity and ionic mobility of the ionogels are relatively lower compared to that in bulk ionic liquid, leading to the narrower electrochemical window. However, we find the CV area of the GQDs//MnO₂ asymmetric MSC is obviously bigger than that of the GQDs//GQDs symmetric MSC at the same scan rate (10 V·s⁻¹) as shown Fig. S5a, suggesting a larger specific capacitance.

In order to confirm the successful construction of GQDs//MnO₂ asymmetric MSCs, the electrochemical properties of the MSCs were evaluated in 0.5 M Na₂SO₄. Fig. 2a shows the characteristics of the asymmetric capacitor, and the CV curves follow a typical rectangular shape even at a very high scan rate of 1000 V s⁻¹ as shown in Fig. 2b, indicating fast charge transfer and fast diffusion of electrolytes in the electrode material.^{5,38} The specific capacitance of the GQDs//MnO₂ MSC is calculated to be 2.98 mF cm⁻² according to GCD curves, and the energy density is 0.414 μWh cm⁻² (with the specific power of 15.01 μW cm⁻² at this point) at the current density of 30 μA cm⁻². The EIS plot is nearly perpendicular to the real axis in the low-frequency region (Fig. 2d), revealing an ideal capacitive behavior. The characteristic frequency (f_0) is the frequency at a phase angle of 45° at which the resistive and capacitive impedance are equal. And the corresponding relaxation time constant ($\tau_0=1/f_0$) is the minimum time needed to discharge all the energy from the device with an efficiency of greater than 50%.^{16,39} The characteristic

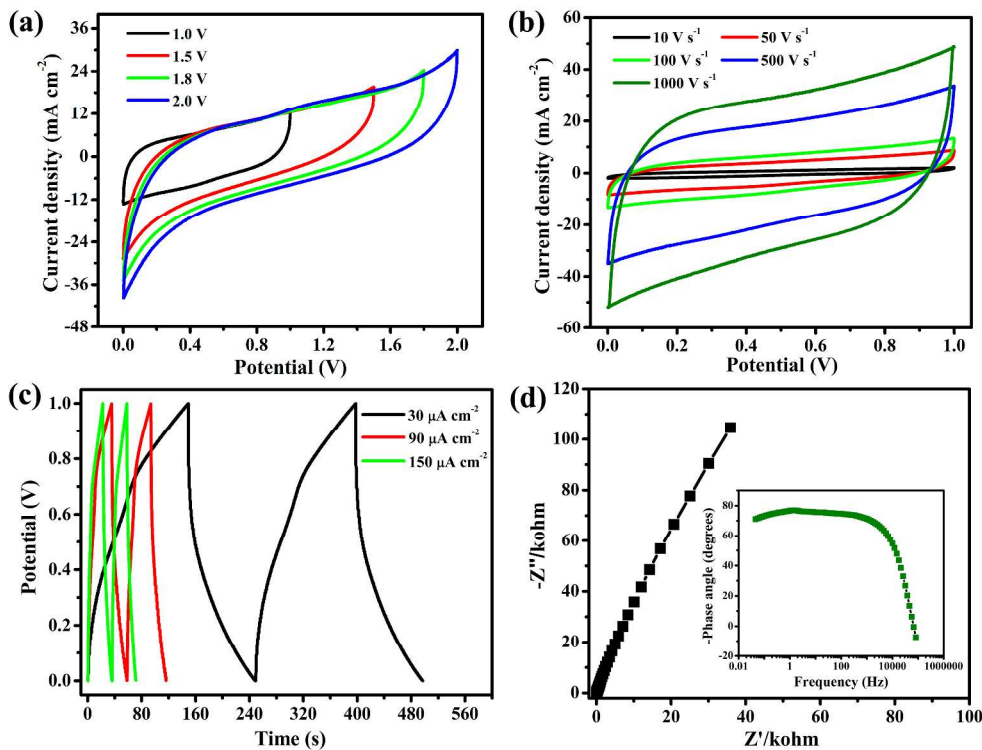


Figure 2. a) CV curves of GQDs//MnO₂ asymmetric MSC with different operation voltages at the scan rate of 100 V.s⁻¹ in 0.5 M Na₂SO₄ electrolyte, b) CV curves of GQDs//MnO₂ asymmetric MSC at different scan rates, c) GCD curves of GQDs//MnO₂ asymmetric MSC at different current densities, d) EIS plot of GQDs//MnO₂ asymmetric MSC, the inset is a plot of the impedance phase angle versus frequency.

frequency of the asymmetric MSC (Fig. 2d inset) is about 15,012 Hz (relaxation time constant $\tau_0=66.6 \mu\text{s}$), demonstrating ultrafast frequency response in the aqueous electrolyte. These results prove that not only was a GQDs//MnO₂ asymmetric MSC successfully constructed, but that the device has excellent electrochemical properties except for the energy density. Next, we will study the influence of different ionogels on electrochemical behavior, and analyze the reason in order to further enhance the energy density of the MSCs.

3.2 Influence of different ionogels on electrochemical behavior of MSCs

Fig. 3 shows the molecular structures of the same cation and two types of anions used to

form ILs, as well as their abbreviations. The chief reason for the choice of $[\text{EMIM}]^+$ as the cation is that EMIM-based ILs have higher conductivity, lower melting points, and lower viscosity comparable to pyridinium cation and long-chain alkane in the imidazole ring, all of which are

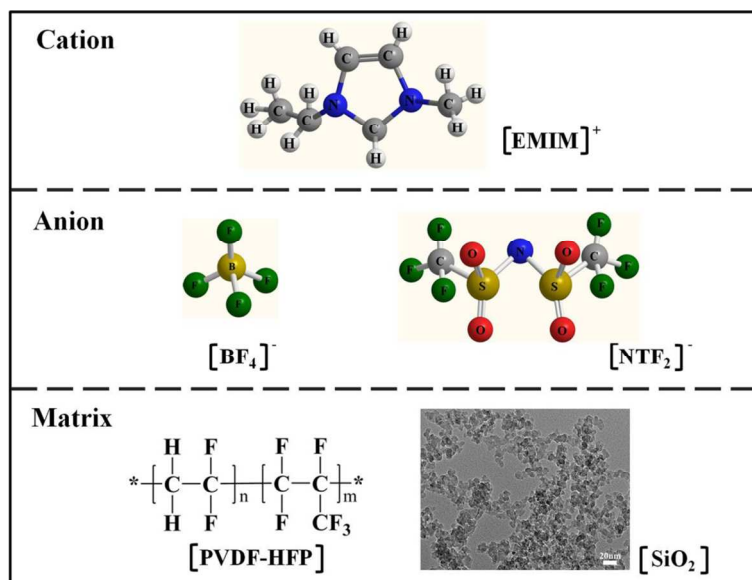


Figure 3. Molecular structures and abbreviations of the cation and two types of anions of ILs, as well as the chemical formula of the polymer matrix and the morphology of nanopowders used to form ionogels.

beneficial to electrochemical properties.⁴⁰ The anions were chosen on the basis of previous results, $[\text{EMIM}][\text{BF}_4]$ and $[\text{EMIM}][\text{NTF}_2]$ ILs have higher conductivity, lower viscosity, and better thermal stability when compared to other anions, with the exception of the for a dicyanamide anion ($[\text{DCA}]^-$). However, although conductivity and viscosity of $[\text{EMIM}][\text{DCA}]$ IL are the best among common ionic liquids, the electrochemical window in $[\text{EMIM}][\text{DCA}]$ is small leading to low energy density.⁴¹ There are two main classes of systems used for immobilizing ionic liquids, organic polymers and inorganic nanopowders (such as SiO_2).⁴² Based on this, we chose the polymer PVDF-HFP and SiO_2 nanopowders as representative matrices.

Based on combinations of the ILs and the matrices, four kinds of ionic liquid gels were ultimately used in this study: EMIM-BF₄-PVDF-HFP, EMIM-BF₄-Fumed SiO₂ (EMIM-BF₄-FS), EMIM-NTF₂-PVDF-HFP, and EMIM-NTF₂-Fumed SiO₂ (EMIM-NTF₂-FS).

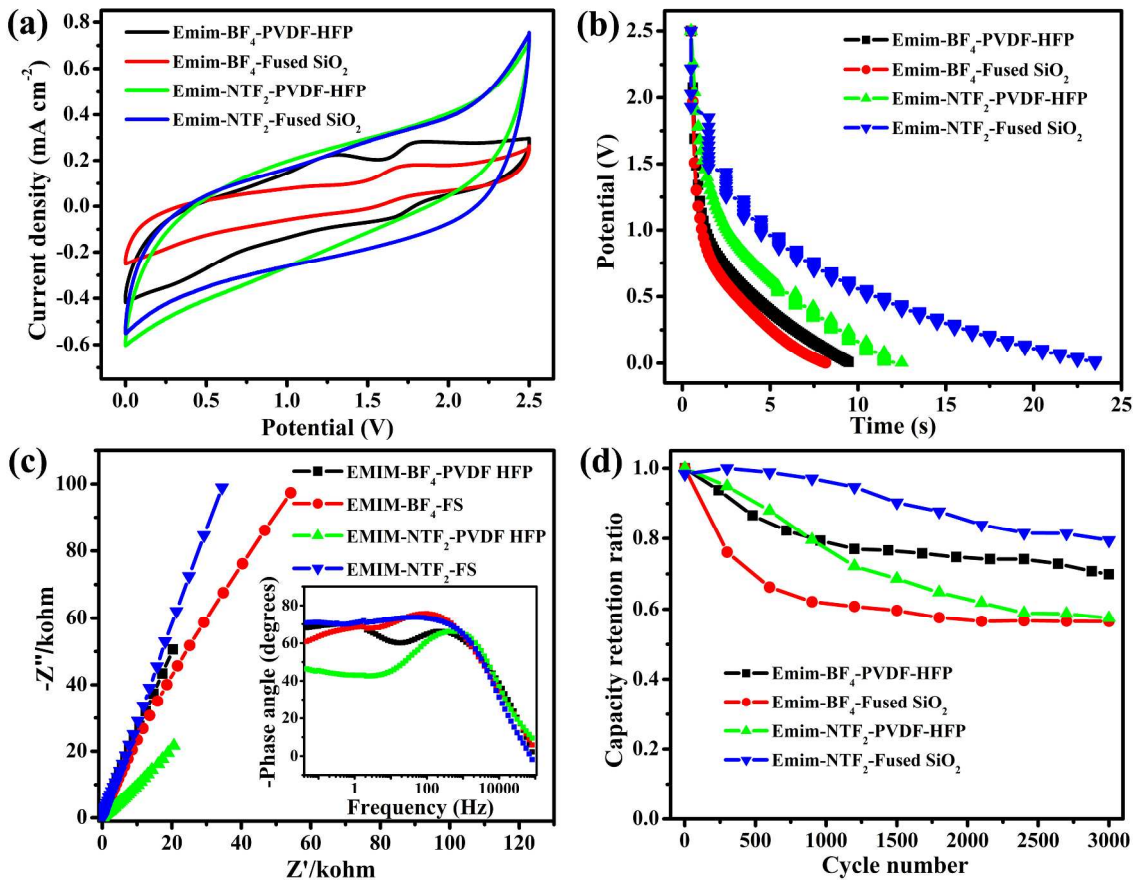


Figure 4. a) CV curves of the GQDs//MnO₂ asymmetric MSCs in four kinds of ionic liquid gels at a scan rate of 1 V s⁻¹, b) GCD discharge curves in the corresponding electrolytes at a current density of 81.5 μ A cm⁻², c) EIS plots measured in the corresponding electrolytes, inset: plot of impedance phase angle versus frequency, d) the cycling stability of asymmetric MSCs in the corresponding ionogels according to CV curves measured at 1 V s⁻¹.

The electrochemical properties of GQDs//MnO₂ asymmetric MSCs in the four ionogels are shown in Fig. 4. CV, GCD and EIS are recognized as the principal methods for evaluating the

fundamental electrochemical behavior of supercapacitors and are shown in Fig4a, Fig 4b, and Fig 4c respectively. The CV curves measured at a scan rate of 1 V s^{-1} exhibit a rapid current response on voltage reversal at the end potential and still remain quasi- rectangular even at the higher scan rate of 10 V s^{-1} and 100 V s^{-1} , as shown in Fig. S6a and b respectively, especially in

Table 1 Equivalent circuit fitting results from the EIS of the asymmetric MSCs in four ionic liquid gels shown in Fig. 4c.

Ionic liquid gels	R_s	$CPE-n$	W
	$[\Omega \cdot \text{cm}^{-2}]$		$[\text{S} \cdot \text{sec}^{0.5} \cdot \text{cm}^{-2}]$
EMIM-BF ₄ -PVDF-HFP	12.93	0.78	1.63E-5
EMIM-BF ₄ -FS	20.19	0.76	4.40E-6
EMIM-NTF ₂ -PVDF-HFP	39.10	0.82	2.29E-5
EMIM-NTF ₂ -FS	14.47	0.89	6.77E-5

the case of EMIM-BF₄-FS and EMIM-NTF₂-FS, indicating ideal exhibit a rapid current response on voltage reversal at the end potential and still remain quasi- rectangular even at the higher scan rate of 10 V s^{-1} and 100 V s^{-1} , as shown in Fig. S6a and b respectively, especially in the case of EMIM-BF₄-FS and EMIM-NTF₂-FS, indicating ideal capacitive behavior.⁴³ The enclosed area of the CV curve in EMIM-NTF₂-FS was found to be larger than that in the other ionogels, suggesting a larger specific capacitance. There are nearly reversible redox peaks existing in the EMIM-BF₄-PVDF-HFP and EMIM-BF₄-FS ionogels, which may be related to storage mechanism of MnO₂ in the ILs consisting of different anions.⁴⁴ The GCD discharge curves (Fig. 4b) also show the highest specific capacity in EMIM-NTF₂-FS, but there is visible IR drops associated with the overall internal resistance of the supercapacitor and relatively low coulombic efficiency (seen from Fig. S6c, d). From the equivalent circuit fitting results shown in Table 1,

the equivalent series areal resistance (R_s) is higher for all of the ionogels than for the aqueous electrolyte due to relatively lower ionic conductivity of the ILs, causing a significant drop in IR. The IR drop can also be partly attributed to the very small relaxation time constant (Fig. 4c) according to the leakage resistance model, $t=\tau_0$ corresponds to the moment that the voltage decreases to 37% of the initial voltage.⁴⁵ The relatively low coulombic efficiency is mainly due to the irreversible capacity resulting from oxygen-containing groups on the surfaces of the GQDs, as demonstrated in our previous research.³⁵ Another probable reason for the low coulombic efficiency is local micro-short circuiting resulting from the short distance (200 μm) between adjacent microelectrodes, which leads to more energy consumption. As can be seen from Fig. 4c and Table 1, each of the EIS plots doesn't show a semicircle at the high frequency region, indicative of fast charge transfer. The straight line in EMIM-NTF₂-FS is most perpendicular to the real axis in the low frequency region, indicating better capacitive behavior, which is in agreement with results in pure IL. The characteristic frequency in EMIM-NTF₂-FS is also highest, demonstrating the fastest frequency response.

Replacing some capacitive circuit elements with a constant phase element *CPE* is often appropriate due to the frequency dispersion of the capacitance. *CPE* behavior is thought to be caused by a distribution of time constants due to inhomogeneous current flow. In the case that $n=1$, a *CPE* simplifies to a capacitor, and when $n=0$, a *CPE* simplifies to a pure resistor.⁴⁶ The value of *CPE*- n is nearest to 1 in EMIM-NTF₂-FS (0.89), indicating most ideal capacitive behavior. The small observed Warburg impedance (W) values can be attributed to the characteristics of [EEMI][NTF₂] as shown in Fig. S7c and its low viscosity (Table S1). It is also possibly a result of interactions between [EMIM][NTF₂] and FS. Asymmetric MSCs using EMIM-NTF₂-FS as the electrolyte have better cycling stability (Fig. 4d), which is related to the

energy storage mechanism in the ionogel and will be discussed later.

In order to further illustrate that asymmetric MSCs with EMIM-NTF₂-FS as the electrolyte have excellent electrochemical performances, we compared the properties in two types of pure ILs (Fig. S7). The CV, GCD, and cycling stability experiments show better electrochemical properties in [EMIM][NTF₂] than that in [EMIM][BF₄]. In the EIS measurement, the straight line is more perpendicular to the real axis in the low-frequency region in [EMIM][NTF₂] (Fig. S7c), revealing an ideal capacitive behavior and better ion diffusion.³⁹ Moreover, several researchers found that the storage mechanism of MnO₂ in ILs is related to the anions of the ILs, and that the anions trigger the pseudocapacitive reaction while the cations are mostly likely just adsorbed/desorbed on the electrode surface.^{44,47} According to our previous research, increasing the number of hydrogen bonds between anions and imidazolium cation leads to higher viscosity as indicated by the ¹H NMR chemical shift of the proton in the 2-position of the imidazole ring.^{41,48} However, low viscosity in [EMIM][NTF₂] results from weak interactions between the anion and cation, which makes reversible insertion/extraction of the anion more easy through the bulk of the MnO₂, contributing significantly to electrochemical properties.⁴⁴ In order to further study the essential properties of the different ionogels to understand the causes of the difference in electrochemical performance, we also measured the ionic conductivity of the ionogels and the temperature dependence of the ionic conductivity.

Ionic conductivity is the most important parameter to evaluate the electrochemical performance of an ionogel electrolyte. Fig. 5 illustrates the temperature dependency of ionic conductivities of the ionogels in this study. According to published literatures and our experiments, we found that EMIM-BF₄-20 wt% FS and EMIM-NTF₂-4 wt% FS are the critical mass fractions to form gel electrolytes, and when the content of PVDF-HFP is less than 20 wt%,

the mechanical strength of the gel electrolytes, EMIM-BF₄-PVDF-HFP and EMIM-NTF₂-PVDF-HFP, noticeably reduce.³³ The experimental data was fitted using the Vogel-Fulcher-Tamman(VFT) equation for ionic conductivity (σ):⁴⁹

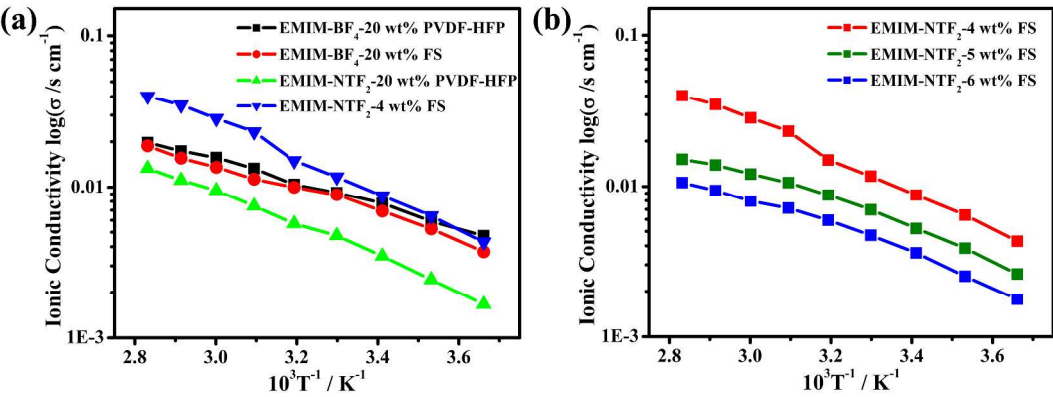


Figure 5. a) Temperature dependence of ionic conductivity of the four kinds of ionic liquid gels with different mass fractions, b) temperature dependence of ionic conductivity for EMIM-NTF₂-FS with different SiO₂ mass fractions.

$$\sigma = \sigma_0 \exp \left[-\frac{B}{(T - T_0)} \right] \quad (5)$$

Where, σ_0 (S cm⁻¹), B (K), and T_0 (K) are constants, and T is absolute temperature. The ionic conductivity value of up to 8.74 mS cm⁻¹ at 20 °C were calculated for EMIM-NTF₂-FS, which is the highest in our as-made ionogels at the same temperature, and can be ascribed to its high molar concentration. For the same concentration, the σ of EMIM-BF₄-PVDF-HFP and EMIM-BF₄-FS is higher than that in EMIM-NTF₂-PVDF-HFP owing to the high conductivity of [EMIM][BF₄]. Fig. 5b shows the temperature dependence of the ionic conductivity of EMIM-NTF₂-FS with different SiO₂ mass fractions. The ionic conductivity of the ionogels discussed

here agrees qualitatively with predictions based on an obstruction model.⁵⁰ Mackie and Meares were the earliest to develop an obstruction expression for solute diffusion in a heterogeneous medium⁵¹

$$\frac{D_g}{D_0} = \left(\frac{1 - \phi}{1 + \phi} \right)^2 \quad (6)$$

Where, D_0 and D_g are the diffusion coefficients of the solute in the bulk state and in the gel, respectively, and ϕ is the volume fraction of the polymer or nanoparticle network. From the Nernst-Einstein equation, we have $\sigma = cz^2 F^2 D / RT$, so we can obtain an analogous expression for σ

$$\frac{\sigma_g}{\sigma_0} = \left(\frac{1 - \phi}{1 + \phi} \right)^2 \quad (7)$$

Where, c is the number of ion carriers, z is the electric charge on each ionic carrier, F is the Faraday constant, R is the gas constant, and T is the absolute temperature. Here σ_0 and σ_g are the ionic conductivities of the ionic liquid in the bulk state and in the gel, respectively.⁵⁰ As shown in Fig. 5b, at high concentrations of [EMIM][NTF₂], the high ionic conductivity of the ionogels is due to the high self-dissociating and ion-transporting abilities of the ionic liquid as well as decoupling of the ion due to interactions between the anions and hydroxyl groups on the surface of fumed SiO₂, which will be further discussed later on.⁵² The electrochemical performance of the GQDs//MnO₂ asymmetric MSC in EMIM-NTF₂-FS (the more suitable electrolyte based on the results above) has been evaluated in detail (vide infra).

3.3 Electrochemical performances of MSC in EMIM-NTF₂-FS and interaction mechanism

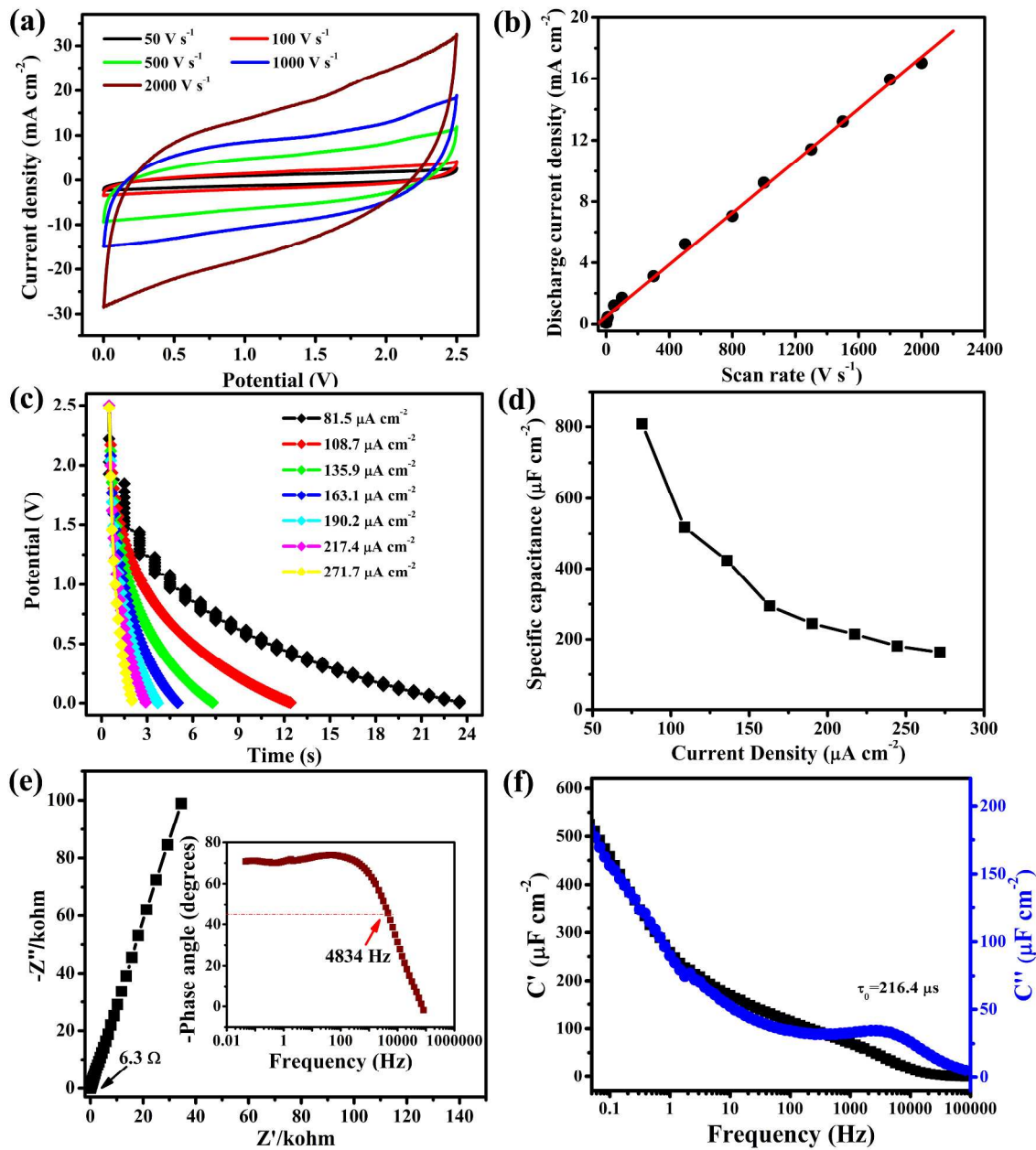


Figure 6. a) CV curves of the GQDs//MnO₂ asymmetric MSC at different scan rates in EMIM-NTF₂-4 wt% Fumed SiO₂ electrolyte, b) The relationship between the charge current densities and the scan rates at 1.25 V, c) GCD curves of the asymmetric MSC at different current densities, d) specific capacitance of asymmetric MSC at different current densities. e) EIS plot of the GQDs//MnO₂ asymmetric MSC, the inset shows a plot of impedance phase angle versus frequency, f) plot of the real or imaginary part (C' or C'') of the specific capacitance versus frequency.

Herein, a GQDs//MnO₂ asymmetric MSC in EMIM-NTF₂-FS (Fig. 6) was successfully fabricated based on the redox characteristics of MnO₂ and the electric double-layer storage of GQDs. As shown in Fig. 6a (from 50 V s⁻¹ to 2000 V s⁻¹) and Fig. S8b (from 0.1 V s⁻¹ to 10 V s⁻¹), the CV curves of GQDs//MnO₂ asymmetric MSC maintain a nearly rectangular CV shape even at an ultrahigh scan rate of 2000 V s⁻¹, indicating ultrafast charging and discharging capability and instantaneous power capability. It deserves be emphasized that a scan rate of 2000 V s⁻¹ is the highest value reported for asymmetric SCs in ionogel system to date.⁷ The linear dependence between the charge current densities and the scan rates is shown in Fig. 6b. A nearly linear increase in the charge current densities for the asymmetric MSC at 1.25 V is observed with scan rates ranging from 0.05 V s⁻¹ to 2000 V s⁻¹, demonstrating a high power output capability and a nearly ideal capacitive behavior. The specific capacitance of the asymmetric MSC is 0.81 mF cm⁻² according to the GCD curves (Fig. 6c, S8b), with an energy density of 0.703 μWh cm⁻² (with a specific power of 101.9 μW cm⁻² at this point) at a current density of 81.5 μA cm⁻². The energy density and power are much higher than those in Na₂SO₄ electrolyte. The IR drop is mainly due to the relatively low ionic conductivity of the ionogel.⁴⁵ The specific capacitance initially drops rapidly (from 808.7 to 214.8 μF cm⁻²) as the current density increases from 81.5 to 217.4 μA cm⁻², but then drops off more slowly at higher current densities. This is most likely related to the pseudocapacitive reaction and storage mechanism of MnO₂ in the ionogel.⁴⁴ The EIS plot reveals an ideal capacitive behavior as well as fast charge transfer. As shown in the inset of Fig. 6e, the frequency in our case reaches 4834 Hz ($\tau_0=1/f_0=206.9$ μs) at a phase angle of -45° (usually used for comparison). To the best of our knowledge, 206.9 μs is the shortest relaxation time constant τ_0 reported in an ionogel system to date, and is significantly shorter than that of high-power SCs fabricated with onion-like carbon (26 ms) in IL.⁵³ In addition, it is better than

those of most of SCs in aqueous electrolytes^{6-8,11,17} and comparable to those of state-of-the-art ultrahigh-power SCs with vertically oriented graphene using 25% KOH as an electrolyte.⁵⁴ Detailed comparisons between the various SCs are summarized in Table 2. In addition, the phase angle value at 120 Hz is usually used as a “factor of merit” to estimate ac line-filtering performance. The impedance phase angle diagram shows that the phase angle reaches up to -73.2° at 120 Hz, making it capable of replacing aluminum electrolytic capacitors (AECs) for the application of 120 Hz filtering.

Table 2. Comparison of the electrochemical performance of various SCs built using different materials

Material	T_0 (ms)	Max scan rate	-Phase at 120 Hz ($^\circ$)	Capacitance (mF cm ⁻²)	Electrolyte
This work	0.21	2000	73.2	0.81	EMIM-NTF ₂ -FS
Electrochemical reduced graphene oxide ⁶	0.24	400	84	0.283	25% KOH
Electrochemically exfoliated graphene and PH1000 ⁷	1~2.9	2000	N/A	0.4~1.8	H ₂ SO ₄ -PVA
Reduced graphene oxide ⁸	1.9	8	75.4	0.32	1 M H ₂ SO ₄
Laser-scribed graphene ¹¹	19	10	<30	N/A	H ₂ SO ₄ -PVA
3D graphene/vertically aligned carbon nanotube arrays ¹⁷	0.82~2.62	500	73.4~81.5	0.23~0.66	1M Na ₂ SO ₄
Onion-like carbon ⁵²	26	200	N/A	N/A	1 M Et ₄ NBF ₄ -PC
Vertical graphene ⁵³	0.067	N/A	82	<0.2	25% KOH

Fig. 6f shows a plot of the real or imaginary part (C' or C'') of the specific capacitance

versus frequency. Complex capacitance modeling of impedance data in relation to frequency is essential for the analysis of supercapacitors.³⁹ Real (C') and complex (C'') capacitance can be expressed in terms of real, imaginary, and total impedance with Eq. 8-11, respectively.⁵⁵

$$Z(\omega) = \frac{1}{j\omega C(\omega)} \quad \text{in ohms} \quad (8)$$

$$C(\omega) = C'(\omega) - jC''(\omega) \quad \text{in farads} \quad (9)$$

Manipulation of Eq. 8 and Eq. 9 leads to

$$C'(\omega) = \frac{-Z''(\omega)}{\omega|Z(\omega)|^2} \quad (10)$$

$$C''(\omega) = \frac{Z'(\omega)}{\omega|Z(\omega)|^2} \quad (11)$$

Where, both C' and C'' have units of farads (F).

C' describes the real part of the capacitance and represents the change in available stored energy as a function of frequency. C'' describes the imaginary part of the capacitance and corresponds to energy loss in the form of energy dissipation. The corresponding relaxation time constant ($\tau_0=216.4 \mu\text{s}$), which is approximately equal to the τ_0 through the impedance phase angle, can also be derived from the frequency at the peak capacitance of C'' . In conclusion, the superior electrochemical performances of GQDs//MnO₂ asymmetric MSCs can be reasonably attributed to synergistic effects between the electrode materials and the ionogel EMIM-NTF₂-FS. Furthermore, the properties of the ionogel EMIM-NTF₂-FS and interaction of [EMIM][NTF₂] and fumed SiO₂ will be investigated next.

Thermogravimetric (TGA) and differential scanning calorimetry (DSC) analyses were carried out at a heating rate of 10 °C min⁻¹ under N₂. For the IL [EMIM][NTF₂] as well as EMIM-NTF₂-FS, the TGA curves indicate degradation of the bulk IL above 320 °C (Fig. 7a). At

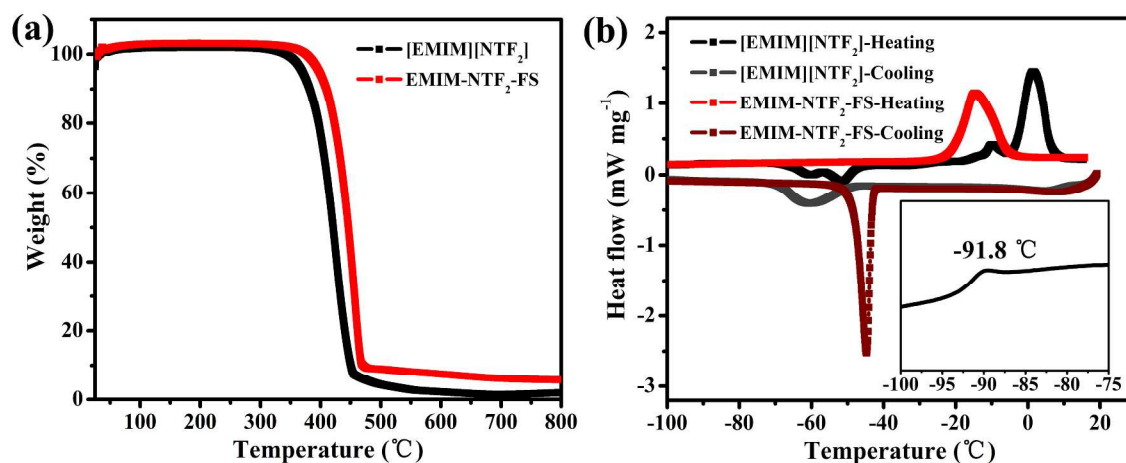


Figure 7. a) TGA curves for the ion liquid and the ionogel EMIM-NTF₂-FS, b) DSC traces of [EMIM][NTF₂] and EMIM-NTF₂-FS during heating and cooling, inset shows the glass-transition temperature of [EMIM][NTF₂].

approximately 500 °C, the IL has fully decomposed. The ionogel EMIM-NTF₂-FS decomposed above 350 °C. It is important to note that about 4 wt% of the material remains relative to the bulk IL, composed of the inorganic silica matrix. The excellent thermal stability of EMIM-NTF₂-FS can be attributed to the confinement effect of fumed SiO₂.⁵⁶ As such, a major advantage of silica-based ionogels is the fact that the thermal stability is limited only by the stability of IL itself, compared to the polymer-based ionogels where the thermal stability can be limited by degradation of the polymer.⁵⁷ DSC measurements of the bulk IL and ionogel gave insight into the confinement effect (Fig. 7b). The DSC of the pristine IL exhibits a glass-transition temperature (T_g) of -91.8 °C (where the compound passed from the glass state to a “subcooled liquid” phase), an exothermic cold crystallization temperature of -60.8 °C (corresponding to the freezing point), and a subsequent endothermic melting point of 1.3 °C. In the ionogels, the temperature of the first-order phase transition (melting) is clearly shifted to a lower value, and the melting point decreases to -14.7 °C. However, the freezing point increases to -44.7 °C.⁴²

These shifts in the phase-transition peaks have been reported for confined fluids.⁵⁸

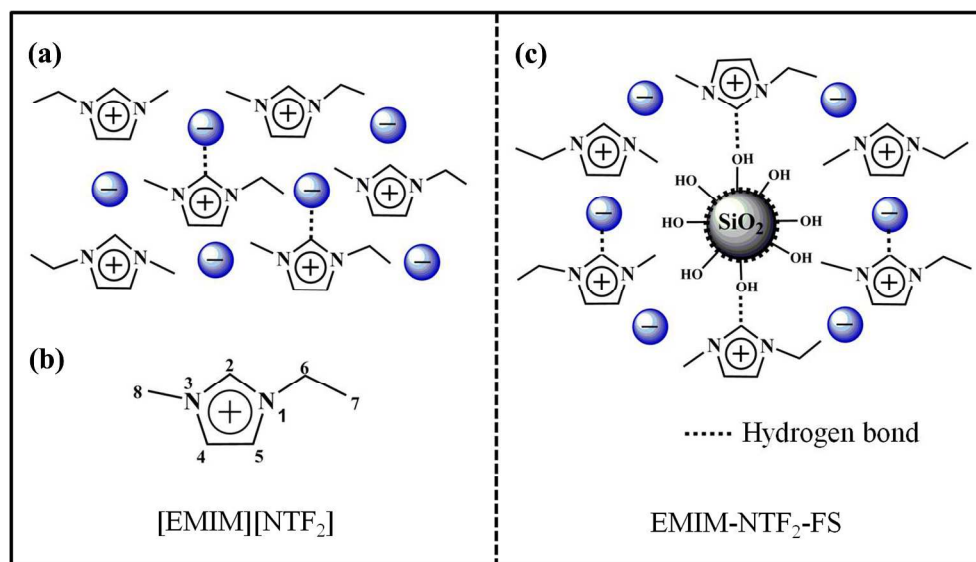


Figure 8. a) Representation of the cation and anion of ionic liquid [EMIM][NTF₂], b) the molecular formula and Cahn-Ingold-Prelog sequence of the cation [EMIM]⁺, c) schematic diagram of the interaction between the ionic liquid and fumed SiO₂.

Based on the characterization of the EMIM-NTF₂-FS gel electrolyte using XPS, FTIR and Raman, a potential mechanism for the interaction between the cation and anion of IL [EMIM][NTF₂] and the matrix and IL is proposed. It is generally well established that hydrogen (H) bonds play an important role in chemistry and biology and that the properties of liquids and solutions are often determined by the strength and number of H bonds.⁵⁹ H bonds, which can form ion-pairs, have been confirmed to be present in ILs using ab initio and DFT methods. In addition, these simulations have revealed that the H atom at the C(2) position is more active in forming H bonds.⁴⁸ Therefore, based on previous literature there is the potential for H bonding C(2)-H...F/O between the C(2)-H of [EMIM]⁺ (Fig. 8b) and F/O atom with higher electronegativities in [NTF₂]⁻ (Fig. 8a).⁶⁰ Moreover, Ludwig et al. also proposed that strong and directional H bonds within ILs could destroy the charge symmetry and thus enhance the fluidity

of ILs, leading to reduced melting points in agreement with DSC results in this study (Fig. 7b).⁵⁹ XPS analysis (Fig. S9) of N 1s and C 1s spectra of EMIM-NTF₂-FS show a low shift of the electron binding energy of the core levels compared with the bulk IL, especially in the N 1s spectra. Specifically, two observable peaks with distinct binding energies can be attributed to the different chemical environments of the N in the IL (Fig. S9a). The N-S bonds in [NTF₂]⁻ have a binding energy of 398.0 eV, while the N-C bond in [EMIM]⁺ has a binding energy of 400.6 eV.⁴⁴ Additionally, the electron binding energy of EMIM-NTF₂-FS in the N 1s spectra is 0.21 eV lower than that in the bulk IL. The broad feature observed at 3433 cm⁻¹ (Fig. S10a), could be due to the formation of more H bonds between the C(2)-H of [EMIM]⁺ and Si-OH groups as shown in Fig. 8b, which would add a small amount of mass in the form of SiO₂ and benefit gel formation. In addition, the electronegativity of the Si-OH groups is introduced into the IL [EMIM][NTF₂], which could increase the electron density of the entire ionogel system. As a result, the core electrons of [EMIM][NTF₂] may experience greater shielding effects, causing a little decrease in the electron binding energy of the core levels, as shown in Fig. S9a and Fig. S9b.⁶¹ All characteristic Raman peaks of [EMIM][NTF₂] can also be observed in the Raman spectrum of ionogel EMIM-NTF₂-FS, except for 475 cm⁻¹ as shown in the inset of Fig. S10b. The band at 475 cm⁻¹ originates from the planar [EMIM]⁺ conformer, which possesses a slightly higher energy (2-4 kJ mol⁻¹).⁶² However, the planar [EMIM]⁺ conformer is presumed to be favored to the adsorbed/desorbed on the MnO₂ surface in comparison with the nonplanar conformer.⁴⁴

4. CONCLUSION

In summary, we prepared all-solid-state asymmetric MSCs on interdigital finger electrodes using GQDs and MnO₂ nanosheets as the negative and positive electrodes, respectively. Based

these devices, we mainly investigated the effect of different ionogels on the electrochemical behavior of MSCs. Our results show that the electrochemical performances of the MSCs using EMIM-NTF₂-4 wt% FS as an electrolyte are much better, presenting high rate capability with a scan rate of up to 2000 V s⁻¹ as well as the fastest frequency response ($\tau_0=206.9\ \mu\text{s}$) recorded to date in an ionogel system. Furthermore, such superior performances of MSCs in EMIM-NTF₂-4 wt% FS are mainly attributed to fast ion diffusion and high ion conductivity. Additionally, the introduction of SiO₂ into the IL allows for the formation of more H bonds between the C(2)-H of [EMIM]⁺ and Si-OH groups, which is beneficial for gel formation. More directional H bonds could also destroy charge symmetry and thus increase the fluidity of the ILs, which can in turn enhance electrochemical performance. In conclusion, our studies enhance the current understanding of the compatibility mechanism of the matrix with ILs, and provide useful information for selecting appropriate ionogel electrolytes for MSCs with desirable properties. Finally, results from this study can aid in the development of high-performance all-solid-state MSCs that will better meet the growing requirements of micro-nano devices for energy storage. And these MSCs in particular have potential applications in ac line-filters.

ASSOCIATED CONTENT

Supporting Information

Structural characterization and optical property of GQDs; schematic illustration of a controllable two-step electro-deposition on an interdigital finger electrode; electrophoretic deposition curve of GQDs; electrochemical deposition curve of MnO₂; CV curves and GCD curves of GQDs//MnO₂ asymmetric MSCs in four ionic liquid gels and in two different pure ILs; viscosity, molecular weight and anion size of two types of ILs; CV curves of GQDs//MnO₂

asymmetric MSCs at low scan rates and GCD curves in EMIM-NTF₂-4 wt%; XPS spectra of the bulk ionic liquid and the corresponding ionogel; FT-IR and Raman spectra of Emim-NTF₂-Fused SiO₂, [EMIM][NTF₂] and fumed SiO₂.

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ACKNOWLEDGEMENTS

This work was supported by Natural Science Foundation of China (No. 21203223).

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